



CONTACT

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PORTFOLIO
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MOBILITY
France, Europe, China,
remote

OBJECTIVE

Seeking a six-month M2 internship from February 2027 in computer graphics, image processing, computer vision, AI, physical simulation, game development or XR.

SKILLS

PROGRAMMING
C, C++, C#, Python, Java

COMPUTER GRAPHICS
OpenGL, GLSL, ray tracing, PBR, real-time rendering, real-time physics simulation, geometry processing

IMAGE PROCESSING
Image coding, segmentation, edge detection, image analysis

AI
Machine learning, foundational deep learning

TOOLS
Git, Linux, Unity, Blender, OpenMP, OpenXR, CUDA

LANGUAGES

FRENCH
Professional working proficiency

ENGLISH
Professional working proficiency

CHINESE
Native

YAQI ZHANG

COMPUTER GRAPHICS | IMAGE | AI | SIMULATION | GAME DEVELOPMENT | VR

Second-year Master's student in Computer Science in the IMAGINE program at Université de Montpellier. I develop interactive systems combining 3D rendering, physical simulation, image processing, artificial intelligence, game development and virtual reality.

EDUCATION

Master in Computer Science - IMAGINE 2025-2027

Université de Montpellier

M1 completed, entering M2. Ranked 3/32, mention bien. Computer graphics, image processing and analysis, artificial intelligence, game development and VR.

Master in Management of Technologies and Sciences 2025-2027

IAE Montpellier - Double degree with the Faculty of Sciences

Complementary training in project management, innovation management, strategy and digital transformation.

DUEF B2 - French Studies 2024-2025

Université de Perpignan Via Domitia

Intensive French program preparing international students for graduate studies in France.

Master's Studies in Software Engineering 09/2022-12/2023

Beihang University

Studied software engineering for approximately one and a half years before changing academic direction and continuing in France.

Bachelor in Software Engineering 2018-2022

Henan University of Technology

Four-year degree covering programming, data structures, systems, networks, databases and software engineering.

SELECTED PROJECTS

NomRigide - PBD Soft-Body Simulator 2026

- Research-oriented TER project in C++/OpenGL for real-time soft-body simulation using Position Based Dynamics.
- Implemented collision constraints, Small Steps, an OpenMP-parallel Jacobi solver and performance analysis tools.

Physicraft - 3D Game Engine 2026

- C++/OpenGL engine with procedural voxel terrain, interactions, PBR rendering and custom rigid-body physics.
- Primarily responsible for rendering and physics: HDR IBL, shadow mapping, collision detection and impulse solving.

CPU Ray Tracer 2025

- C++ renderer supporting reflection, refraction, soft shadows, textures and triangle meshes.
- Kd-Tree acceleration, OpenMP parallel rendering and MSAA.

GJ Fly - 2D Platformer, 3X-Schools GameJam 2023

- Unity/C# multi-level platformer with jumping, dashing, combat, enemies and environmental hazards.
- Programmed the player state machine, level interactions, projectiles and object pooling.

EXPERIENCE

Virtual Reality Laboratory - Beihang University 04/2023-08/2023

- Studied computer graphics and VR application development in my academic advisor's laboratory.
- Contributed to a VR spatial-navigation assessment system led by the Faculty of Psychology at Beijing Normal University.

JingHang Telek Technology, Beijing - Java Full-Stack Developer | Full-time 07/2022-02/2023

- Contributed to a line-loss analysis system for State Grid Jiangxi.
- Developed core backend functionality and clear, user-friendly interfaces.